A Robust Discrete-Time Controller for DC-AC Inverters

Gustavo Willmann
Dept. of Engineering and Development - CP Eletrônica
Rua da Várzea, 379
Porto Alegre/RS, 91040-600
BRAZIL
willmann@cp.com.br

Daniel F. Coutinho, Luís F. A. Pereira
Department of Electrical Engineering
Pontifícia Universidade Católica do Rio Grande do Sul
Av. Ipiranga, 6681, Porto Alegre/RS
90619-900, BRAZIL
daniel@ee.pucrs.br, pereira@ee.pucrs.br

***Abstract* – This paper presents a robust control strategy for DC-AC power inverters based on the linear matrix inequality (LMI) framework. More precisely, a constant discrete-time state feedback controller is proposed taking into account the average model. The strategy assures the local stability of the closed-loop system for all admissible linear load variation while minimizing the effects of nonlinear loads. Practical results show the advantages of the approach over classical control design techniques.**

I. INTRODUCTION

The design of controllers to dc-ac power converters such as uninterruptible power supplies (UPS) and automatic voltage regulators (AVR) has aroused much attention from the control community due to the wide variety of applications in industrial environments. However, load variations (generally, unknown but bounded), exogenous signals, control signal constraints and nonlinear behavior of switched systems make difficult the task of designing controllers whenever some kind of performance is desired to the closed-loop system [1]. For industrial applications, the power converters are generally controlled by means of dual loop leg-lead compensators (e.g., PID controllers) using the average model of the switched system governed by the PWM technique [2,3]. Frequency-domain techniques have several constraints to designing multi-loop control systems that are addressed by considering a decoupling between fast and slow dynamics leading to a poor static regulation and bad dynamical response (mainly, when an abrupt change on the load value occurs). Besides, there exist heavy harmonic components on the output signal whenever the UPS is supplying energy to nonlinear loads.

Due to the appearance of new promising control approaches, associated to the dissemination of powerful digital signal processors devoted to control applications, many researchers have applied a wide diversity of techniques aiming the improvement on static and dynamic performance of dc-ac power converters. Among this line, one can cite the dead-beat controller of Jung et al in [4], the sliding mode control of Tai and Chen [5], and the model reference control of Rech et al in [6]. In parallel, exploiting the uncertain nature of the output-load, some researchers have applied the robust control theory of linear time-invariant (LTI) to control UPS systems such as the references [7-10].

On the other hand, the robust control technique based on linear matrix inequalities (LMIs) is being applied to solving a large diversity of control problems, see e.g. [11]. The LMI framework can deal with control constraints such as robustness, disturbance rejection and bounded control signals using the same foundations. For instance, the stabilization, performance design and control constraints are added to the same formulation and then solved in one single optimization problem. Basically, modified Lyapunov inequalities (including the control problem and performance constraints) are recast as a set of LMIs, which are numerically solved by means of standard packages [12].

In this paper, a discrete-time robust state feedback control law is designed to the ac-voltage regulation of a power converter that is subject to uncertainty (a linear load admittance variation), disturbance signals (which may represent a nonlinear load) and also bounded control signals (actuator saturation). Differently, from other approaches, the linear load may vary free in time with known (maximal and minimal) limits, and the initial conditions are unknown but belonging to a given ellipsoidal set. The proposed approach is then applied to a commercial power converter where the control strategy is implemented using a real-time Matlab-based platform. To test the overall performance, a comparative study is carried out with an industrial multiple-loop PID controller having a similar structure of the proposed strategy.

II. MATHEMATICAL MODEL

Basically, the inverter is composed by a half-bridge DC-AC inverter with a standard LC filter. The load voltage is the variable to be controlled by means of the PWM technique. In this case, the following simplified average (time-varying) model of the power converter will be considered in this paper:

$$\begin{aligned}\dot{x}(t) &= A(Y_O(t))x(t) + Bu(t) + Ei_o(t),\\ z(t) &= Cx(t) + Du(t)\end{aligned} \tag{1}$$

where $x(t)=[x_1(t)\ x_2(t)]^T$ is the state vector, where $x_1(t)$ stands for the inductor current, and $x_2(t)$ is the capacitor voltage, $u(t)$ is the PWM signal, $i_o(t)$ is the output disturbance assumed to have a bounded energy (possibly the current driven by a nonlinear load) and $z(t)$ represents the variables of interest (e.g., the output voltage). The state matrix $A(Y_O(t))$ is an affine matrix function of the load admittance which means that the load is uncertain with an unknown rate of variation, but bounded by:

$$Y_{\min} \le Y_O \le Y_{\max} \tag{2}$$

For simplicity of notation, the above relationship is denoted as follows:

$$Y_O \in \Theta = \{Y_O : Y_{\min} \le Y_O \le Y_{\max}\} \tag{3}$$

Using elementary electric circuit theory results in the following state space characterization:

$$A(Yo)=\begin{bmatrix}-\left(\dfrac{R_{Lf}}{Lf}\right) & \dfrac{-1}{Lf}\\ \dfrac{1}{Cf} & \dfrac{-Yo}{Cf}\end{bmatrix},\ B=\begin{bmatrix}\dfrac{1}{Lf}\\ 0\end{bmatrix},E=\begin{bmatrix}0\\ \dfrac{-1}{Cf}\end{bmatrix},$$

where R_{lf} is the equivalent inductor resistance, Y_O is the load admittance (suppose to be time-varying) and Lf and Cf are respectively the output filter inductance and capacitance.

Aiming a discrete-time implementation, the state variables are measured and then sampled using a standard ZOH at a sampling rate T_s. So, a discrete-time state representation may be obtained by means of the Euler approximation technique, i.e.

$$\frac{dx(t)}{dt}\cong\frac{x(t+T_s)-x(t)}{T_s},$$

resulting in the following dynamics:

$$x(k+1)=A_d(Y_o)x(k)+B_d u(k)+E_d i_o(k), \tag{4}$$

where $A_d(Y_o)=(I+T_sA(Y_o))$, $B_d = T_sB$ and $E_d = T_sE$.

III. CONTROL DESIGN

The problem to be addressed in this section is to design a control law for system (4) that tracks a sinusoidal reference voltage $v_{ref}(k)$. To this end, we redefine the state $x_2(k)$ as the tracking error, i.e.,

$$x_2(k)=v_{ref}(k)-v_o(k),$$

and augment the original system using a discrete-time integrator adding the following dynamics

$$x_3(k+1)=T_s x_2(k)+x_3(k).$$

The above manipulations lead to the following discrete-time representation:

$$x(k+1)=A_D(Y_o)x(k)+B_D u(k)+E_D(Y_o)w(k), \tag{5}$$

where the matrices and vectors are given by:

$$x(k)=[x_1(k)\quad x_2(k)\quad x_3(k)]^T,\ w(k)=[i_o(k)\quad v_{ref}(k)]^T$$

$$A_D=\begin{bmatrix}\left(1-\dfrac{T_S R_{lf}}{L_f}\right) & \dfrac{T_s}{L_f} & 0\\ -\dfrac{T_s}{C_f} & \left(1-\dfrac{T_sY_o}{C_f}\right) & 0\\ 0 & T_s & 1\end{bmatrix}, E_D=\begin{bmatrix}0 & -\dfrac{T_s}{L_f}\\ \dfrac{T_S}{C_f} & \dfrac{T_sY_o}{C_f}\\ 0 & 0\end{bmatrix},$$

$B_D=\begin{bmatrix}\dfrac{T_s}{L_f} & 0 & 0\end{bmatrix}^T$, and the superscript symbol ^{T} denotes the transpose operation.

The control objective is to design a state feedback gain K, i.e. $u(k)= K\,x(k)$, such that the system defined in (5) is stable for all admissible Y_o while requiring the following:

i. Minimize an upper-bound on the L_2-gain of the input-to-output operator. That is, from the disturbance input $w(k)$ to the performance signal $z(k)$, we minimize a scalar γ which bounds the following:

$$\|G_{wz}\|_\infty=\sup_{0\neq w}\frac{\|z(k)\|_2}{\|w(k)\|_2}<\gamma \tag{6}$$

ii. The control signal should be constrained to the following set:

$$U=\{u(k):|Kx(k)|\le u_{\max}\}$$

whenever the initial conditions belong to the following ellipsoidal set:

$$X_0=\{x(0):x(0)^T P_0 x(0)\le 1, P_0=P_0^T>0\}$$

iii. The rate of convergence of the state trajectory is bounded by a given constant r.

The design specifications can be described in the LMI framework by means of the modified Lyapunov inequalities, i.e. the Bounded Real Lemma [15]. To make this point clear, an upper-bound on the L_2-gain, as defined above, can be determined by minimizing γ subject to:

$$\exists V(x)>0: r\Delta V(x)+\frac{1}{\gamma}z^T z-\gamma w^T w<0 \tag{7}$$

for all $Y_O\in\Theta$, where $V(x)$ is a Lyapunov function that proves the stability of system (5) in closed-loop, $\Delta V(x)$ stands for the Lyapunov function variation, i.e.

$$\Delta V(x)=V(x(k+1))-V(x(k)),$$

and r the rate of convergence.

We have assumed that the load admittance Y_O is uncertain varying free in time, so the quadratic function as stated below will be considered as the Lyapunov function candidate:

$$V(x(k))=x^T(k)Px(k), \tag{8}$$

where P is a (constant) symmetric positive definite matrix to be determined.

A. LMI Conditions for H_∞ Control Design

The LMI-based methods allow the consideration of a wide diversity of constraints in a unified framework. For instance, one can rewrite the condition (7) in terms of LMIs to guarantee the robust stability and the performance requirement (i). The remaining ones can also be recast as new LMI constraints that are added to the original optimization problem.

To make this point clear the inequality in (7) taking into account (8) can be written as follows:

$$\begin{bmatrix}x(k)\\ w(k)\end{bmatrix}^T\begin{bmatrix}A_{mf}^T rPA_{mf}-rP+\dfrac{1}{\gamma}C_{mf}^T C_{mf} & A_{mf}^T PE_D\\ E_D^T PA_{mf} & E_D^T PE_D-\gamma I\end{bmatrix}\begin{bmatrix}x(k)\\ w(k)\end{bmatrix}<0$$

where $A_{mf} = A_D + B_D K$, $C_{mf} = C + D K$, and r is a scalar defining the convergence rate of the state trajectory [14], i.e.

$$\|x(k)\|\le\sqrt{\lambda_2/\lambda_1}(r)^k\|x(0)\|,$$

where λ_1 and λ_2 are respectively the smallest and largest eigenvalues of P.

From standard LMI machinery (see [11,12] for further details), the above is satisfied for a given K and all $Y_O\in\Theta$ if the following holds:

$$\begin{bmatrix} -rP & 0 & A_{mf}^T P & C_{mf}^T \\ 0 & -\gamma I_2 & E_D^T P & 0 \\ PA_{mf} & PE_D & -rP & 0 \\ C_{mf} & 0 & 0 & -\gamma \end{bmatrix} < 0, Y_0 \in \Lambda(\Theta),$$

where $\Lambda(\Theta)$ stands for the set of vertices of Θ (i.e., $\Lambda(\Theta)$ = $\{\Upsilon_{min}, \Upsilon_{max}\}$).

The above condition is an LMI whenever the control gain is given. However, one can find a solution in K and P rewriting the condition in terms of the Lyapunov matrix inverse, $Q = P^{-1}$, and using the parameterization $L = K\,Q$ [12].

From the above analysis, condition (7) is satisfied if there exist matrices Q (symmetric) and L, and a positive scalar γ solving the following optimization problem for all $Y_0 \in \Lambda(\Theta)$:

min γ subject to

$$\begin{bmatrix} -rQ & 0 & QA_D^T + L^T B_D^T & QC^T + L^T D^T \\ 0 & -\gamma I_2 & E_D^T & 0 \\ A_D Q + B_D L & E_D & -rQ & 0 \\ CQ + DL & 0 & 0 & -\gamma \end{bmatrix} < 0 \quad (9)$$

In affirmative case, the closed-loop system in (5) with

$$u(k) = (LQ^{-1})x(k) \quad (10)$$

is robustly stable with respect to all admissible Y_0 and

$$V(x(k)) = x^T(k)Q^{-1}x(k)$$

is a Lyapunov function that proves it. Moreover, the L_2-gain of the closed-loop system satisfies (6) and its rate of convergence is lower than r.

B. Control Signal Constraint

The control effort constraint in (ii) can be established inside the set X_0 containing all initial conditions of system (5). Normally, the set of initial conditions is over-bounding by the following region:

$$R_A = \{x(k) : V(x(k)) = x^T(k)Q^{-1}x(k) < 1\}, \quad (11)$$

that we hereafter will denote as the domain of non-saturation of system (5). In other words, $X_0 \subset R_A$. The inclusion condition can be cast as an LMI constraint as follows:

$$\begin{bmatrix} P_0 & I_2 \\ I_2 & Q \end{bmatrix} > 0 \cdot \quad (12)$$

If it is required that the initial condition $x(0)$ of system (1) does not drive the control signal outside the region R_A then the following condition must be satisfied:

$$R_A \subset U = \{x(k) : |Kx(k)| \le u_{max}\} \quad (13)$$

The above condition can be rewritten as follows:

$$1 - x^T Q^{-1} x > 0, \forall\, x : |Kx| - u_{max} \le 0 \quad (14)$$

Applying the technique known as S-Procedure [12], one can write the above condition as a new LMI constraint. More precisely, we get the following additional constraint:

$$\begin{bmatrix} u_{max}^2 & L \\ L^T & Q \end{bmatrix} > 0 \quad (15)$$

To sum-up, one can obtain a solution to (i)-(iv) if there exists a solution to the following optimization problem:

$$\min_{Q,L,Y_0 \in \Lambda(\Theta)} \gamma : (9), (12), (15).$$

IV. EXPERIMENTAL RESULTS

A real-time platform composed by a commercial DC-AC inverter and a data acquiring card is used to analyze the performance of the proposed approach. Also, for comparison purposes, the same tests are applied to the original multi-loop PID controller used in the power converter.

The inverter parameters are detailed below:

- Inductance Lf = 1 mH;
- Capacitance Cf = 300 μF;
- DC source, 180 V (360 peak-to-peak);
- IGBT Semikron SKM 300 GB 063D;
- 3φ rectifier Crydom M5060TB1200;
- *Driver* Semikron SKHI22A;
- A reference voltage (sinusoidal signal) with a frequency of 60 Hz and an amplitude of $110V_{RMS}$

A. Real-time Plataform

The controller action and measurement is performed by a data acquiring card connected to a PC computer, running a real-time application in *Matlab*®.

The DC voltage source is the one presented in Figure 1, where its components are:

- C1 and C2: 6800 μF / 350 V;
- D1 to D6: rectifier diodes;
- V_r, V_s, V_t is a 3φ voltage-source with $127V_{RMS}$.

The single phase inverter can supply to the load up to 5KVA. It turns out that the methodology can be applied to more powerful converters, even for the 3-phase ones, since we have only to replace the LC and load values.

The data card used in this implementation is the PCI-DAS1200 from *Computer Boards*. The card contains multiples (analog and digital) input-output channels as described below:

- 8 differential or 16 standard analog inputs with a resolution of 12-bits;
- 24 digital inputs that can be used as inputs or outputs both being fully programmed;
- 2 outputs with a 12-*bit* D/A converters with programmable gains and DC levels.

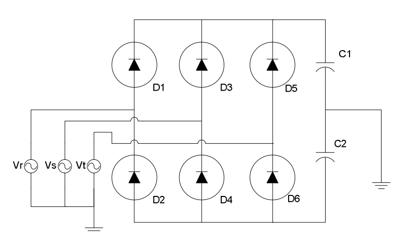

Figure 1 – DC bus.

The total shared sampling rate is over 330 KHz. The data card is controlled by means of a real-time application running over the *Matlab*® 6.5 R13 with a supplied driver for both *Real-*

Time Windows Target and *xPC Target Simulink* environments. The computer in which the card is installed is a Pentium 4, 2.4 GHz, with 256MB of RAM.

Figure 2 shows a schematic diagram of a *Simulink®* applicative used as an interface between Matlab® and the data acquisition board.

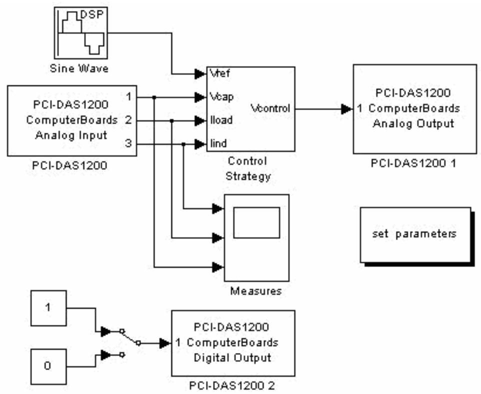

Figure 2 – Simulink environment for practical implementation.

B. RMS Value Computation

To regulate the output voltage in 110 V_{RMS}, we have considered a double loop structure (inner and outer loops). The internal loop is responsible for the sine wave tracking and to assure a nice dynamical response, as shown in Figure 3.

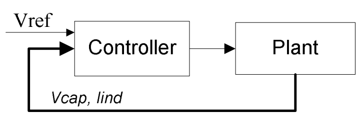

Figure 3 – Internal control-loop.

A second and external loop is added to the control structure to compensate steady-state errors in the RMS output voltage due to the DC link variation and the LC filter losses whenever there is an output perturbation (load variation). To this end, the RMS value of the output voltage is computed and then compared with the RMS reference voltage to produce a correcting signal that modulates de amplitude of reference sine wave. The whole structure of the double-loop control strategy is illustrated in Figure 4.

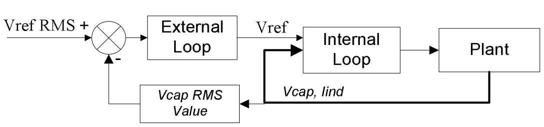

Figure 4 – Multiple-loop control strategy.

C. PID Controller

The PID control strategy is largely used in industrial applications due to its simple (analog and digital) implementation and a vast literature on tuning its parameters. To contrast with the proposed methodology, the original double-loop PID controller used in the inverter is considered. The PID controller was designed using standard frequency-domain techniques [15], and then its performance was improved by exhaustive tests in laboratory to achieve the best combination between transient performance and low harmonic distortion.

The multi-loop structure is similar to the one considered in [3] where the capacitor voltage and current are used for feedback. Due to commercial reasoning, only the capacitor voltage is measured and the current is estimated by means of a derivative action. For a best understanding, the inner loop of the original controller of the power converter is shown in Figure 5.

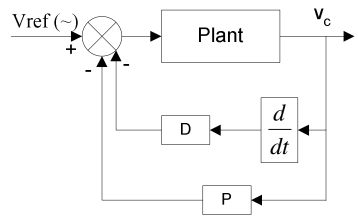

Figure 5 – Inner-loop of the PID implementation.

Basically, the first loop, a PD action, aims to produce a good dynamic response for a sinusoidal tracking. The external loop, a PI action, improves the RMS voltage regulation using the same structure illustrated in Figure 5. The *Simulink* block that implements the two-loop PID is described in Figure 6.

The PID gains used in the experimental tests are given by:

- Internal-loop: $P = 1.5$ and $D = 3.5$;
- External-loop: $P = 0$ and $I = 0,6$.

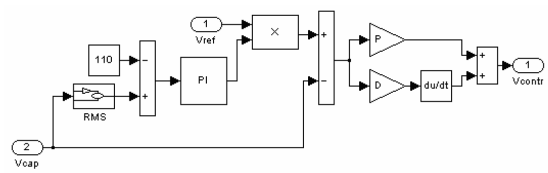

Figure 6 – Two-loop PID implementation.

To analyze the closed-loop performance, we have applied a step load that consists of a rectifier stage (nonlinear load), as shown in Figure 7, where $C1 = 2200\ \mu F$ and $R1 = 5\Omega$, resulting in a 0.7 power factor. Besides, a regulation of 10% is considered on the DC bus whenever the rated nonlinear load is connected to the output. The experimental test includes both a load step and a load disconnection step.

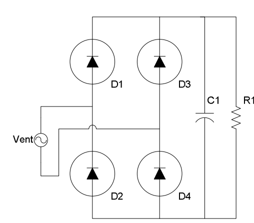

Figure 7 – The nonlinear load.

Figure 8 shows the instantaneous and RMS value of the output voltage in which a step on the nonlinear load is applied.

Notice that a slower response on the output voltage appears due to the integral action on the external control-loop, the deviation is *6 V*olts and the settling-time (for reaching the RMS reference) is more than *100 ms*.

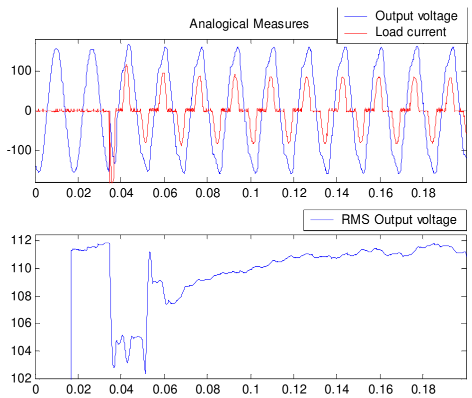

Figure 8 – PID performance for nonlinear load step variation.

The transient response of the DC-AC inverter controlled by a 2-loop when considering that the nonlinear load is disconnected from the output is showed in Figure 9. In this case, a *10 Volts* over-shoot has appeared on the RMS output voltage due to the integral action (and a possible wind-up effect) and taking approximately *90 ms* to return to the reference value.

Table 1: PID controller – THD analysis.

Voltage	THD	Current	THD
111.0V	1,0%	0A	0
110.7V	10,8%	41A	65%

As expected, the PID controller has high harmonic distortion (see Table 1), since we have not considered an additional-loop using the output current information to reduce the THD as in [4] where $i_0(k)$ is taken into account to correct the inductor current value. We stress the fact that the main objective in this work (a first experimental implementation) is to compare the transient performance of a standard controller used in an industrial power inverter system and the proposed approach.

D. Robust Control Strategy

It turns out that the proposed approach has a similar feedback structure of the PID strategy, only differing by considering the inductor current for feedback instead of the (estimate of the) capacitor current as in the multi-loop PID. Notice that the i_C is linearly dependent on the state-variables and the disturbance signal as below

$$i_C = x_1 - Y_0 x_2 - i_0 ,$$

and so the same control action can be achieved by redefining adequately the control gains. To illustrate the above statement, Figure 10 shows the inner-loop of the proposed strategy that can be compared with Figure 5.

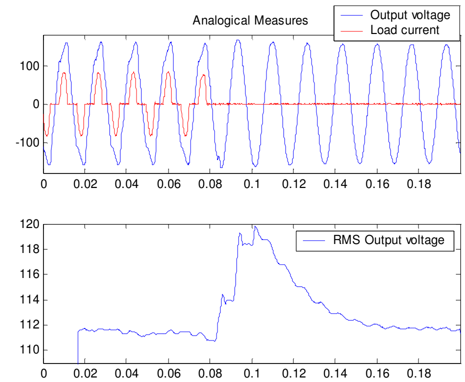

Figure 9 – PID performance for a nonlinear load disconnection.

To design the control gains by means of the optimization problem stated at the end of Section III, the following parameters from the industrial DC-AC inverter are considered:

- $Y_{MIN} = 0.01$ and $Y_{MAX} = 0.2$ - the admissible minimum and maximum load of the inverter;
- $u_{MAX} = 180V$ – the DC-link voltage;
- $r = 0.8$ – the best tradeoff between transient response and voltage regulation; and
- $z(k) = x_2(k)$.

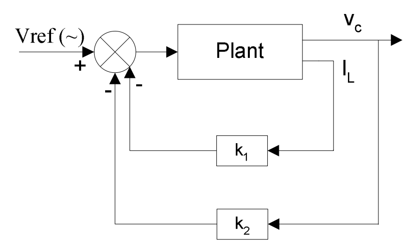

Figure 10 – Inner-loop of the proposed approach.

Applying the LMI toolbox of *Matlab*® yields the following control gains:

- $K_1 = -6.0,\ K_2 = -8.0,\ K_3 = 3.0$.

The proposed control strategy is implemented by means of the *Simulink* block described in Figure 11.

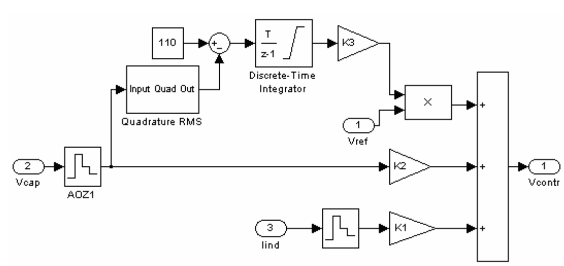

Figure 11 – The robust control implementation.

Figure 12 shows the output voltage (sine wave and RMS value) response of the proposed controller considering the same conditions of the PID strategy (a nonlinear load step).

The proposed controller has obtained a settling time of less than *60ms* and a voltage deviation of *25.0 Volts*.

Similarly to the PID method, the nonlinear load is disconnected and the output voltage response is showed in Figure 13. The settling-time is approximately *70 ms* with an over-shoot of *18.0 Volts*.

Table 2 shows the harmonic distortion of the output voltage and current supplied by the UPS with approximately the same levels of the PID controller.

Table 2: harmonic distortion of the proposed controller.

Voltage	THD	Current	THD
110.1V	0,5%	0A	0
109.5V	11,7%	39A	65%

We stress the fact that both controllers have basically the same structure and so they might behave in a similar way if suitable control-gains are setting. The major advantage of the proposed approach is to give a (sub-) optimal solution regarding the design constraints inherent of practical applications in a single optimization problem solved in a numerical and tractable way avoiding try and error methods.

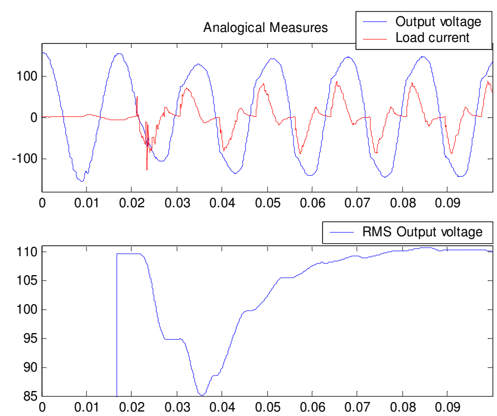

Figure 12 – Controller performance for a nonlinear load.

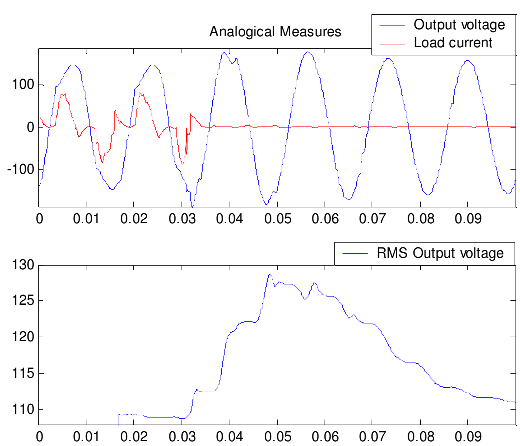

Figure 13 - Controller performance for load disconnection.

V. CONCLUSION AND FUTURE WORK

This paper has proposed a robust state-feedback strategy to the control of DC-AC power converter supposing several constraints such as an uncertain linear load, nonlinear load disturbances, among others. The experimental results have shown a better dynamic response of the proposed methodology over the standard PID controller. It turns out that the proposed controller has been designed in a single optimization step with guaranteed performance contrasting with standard control synthesis techniques that needs an extra heuristic fine tuning. A methodology to define the optimal value of the rate of convergence is being carried out by the authors.

REFERENCES

[1] G. Escobar, R. Ortega, H. Sira-Ramírez, J.-P. Vilain, and I. Zein, "An Experimental Comparison of Several Nonlinear Controllers for Power Converters", *Cont. Syst. Magazine*, pp. 66-82, 1999.

[2] N. Mohan, T. M. Undeland, W. P. Robbins, *Power Electronics: converters, applications, and design*, John Wiley & Sons, 2nd Edition, 1995.

[3] N. M. Abdel-Rahim, and J. E. Quaicoe, "Analysis and Design of a Multiple Feedback Loop Control Strategy for Single-Phase Voltage-Source UPS Inverters", *IEEE Trans. Power Elect.,* vol. 11, pp. 532-541, 1996.

[4] S.-L. Jung, H.-S. Huang, M.-Y. Chang and Y.-Y. Tzou, "DSP-Based Multiple-Loop Control Strategy for Single-Phase Inverters Used in AC Power Sources", *in Proc. 28th Annual IEEE Power Elect. Spec. Confer.,* vol. 1, pp. 706-712, 1997.

[5] T. Tai, and J. Chen, "UPS Inverter Design Using Discrete-Time Sliding-Mode Control Scheme", *IEEE Trans. Ind. Elec.*, v.49, no.1, pp. 67-75, 2002.

[6] H. Pinheiro, A. S. Martins and J. R. Pinheiro, "A Sliding Mode Controller in Single Phase Voltage Source Inverters", *in Proc. 20th Int'l Conf. Ind. Elect., Contr. Instrum.,* vol. 1, pp. 394-398, 1994.

[7] V. F. Montagner, and P. L. D. Peres, "Robust State Feedback Control Applied to an UPS System". *In Proc. 29th Annual Conf. IEEE Ind. Elect. Soc.*, vol. 3, pp. 2245-2250, 2003.

[8] R. Naim, G. Weiss, and S. Ben-Yaakov, "H_∞ Control Applied to Boost Power Converters", *IEEE Trans. Power Elect.*, vol. 12, 677-683, 1997.

[9] T.-S. Lee, S.-J. Chiang, and J.-M. Chang, "H_∞ Loop-Shaping Controller Designs for the Single-Phase UPS Inverters", *IEEE Trans. Power Elect.,* vol. 16, pp. 473-481, 2001.

[10] T.-S. Lee, K.-S. Tzeng, and M.-s. Chong, "Robust controller design for a single-phase UPS inverter using μ-synthesis", *IEE Proc.-Electr. Power Appl.,* vol. 151, pp. 334-340, 2004.

[11] L. El Ghaoui and S-I. Niculescu, Advances in Linear Matrix Inequality Methods in Control, SIAM books, 2000.

[12] S.P. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in Systems and Control Theory*, SIAM books, 1994

[13] J. B. Burl, *Linear Optimal Control*, Addison-Wesley, 1999.

[14] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*, Prentice Hall, 1991.

[15] G. F. Franklin, J. D. Powell, and A. Emani-Naeini, *Feedback Control of Dynamic Systems,* Addison-Wesley Publishing Company, 1994